SYNAPSE EDGE DEFENSE

Request Processing Lifecycle

Complete request lifecycle — identity, inspection, profiling, scoring, decision. All intelligence runs locally at the edge. No network hops. No backend calls.

<50µs	29–71µs	~450µs	0
WAF DETECTION	DLP SCAN	E2E PROXY	NETWORK HOPS

BLOCKING PATH — SEQUENTIAL PIPELINE

<50µs detection latency

IN	01	02	03	04	05
Request	**TLS Fingerprint**	**WAF Detection**	**API Profiling**	**Actor Scoring**	**Decision**
HTTP/HTTPS Headers + Body	JA4 / JA4H Client identity	237+ Rules SQLi/XSS/RCE	Schema + Anomaly ~5µs validation	Risk 0–100 History + campaigns	Allow / Block Threshold: 70

ASYNC — PARALLEL: DLP Scanning — 22+ sensitive data patterns · PII, API keys, credentials · Aho-Corasick matching · Runs during response body streaming — zero added latency

STAGE DETAILS

No stage requires network calls or external dependencies

TLS Fingerprinting — ~1µs

JA4 (TLS) + JA4H (HTTP) hashing
Detect spoofed User-Agents
Track actors across IP rotation

→ Reputation tracking per fingerprint

WAF Detection Engine — 15–25µs

237+ production rules (libsynapse)
30+ match types (SQLi, XSS, RCE...)
Hot reload without restart

→ Attack detection: 15–25µs typical

API Profiling — ~5µs

Schema learning (~5µs validation)
Endpoint baselines (rate, size, headers)
Anomaly detection (statistical deviation)

→ Flags zero-day attacks by behavior

Actor Scoring — ~1µs

100K actor cache (IP + JA4 + token)
Risk decay: 10 pts/min (configurable)
Campaign membership detection

→ 7 correlation detectors identify coordinated attacks

Decision Engine — <1µs

Threshold: 70 = block (configurable)
Actions: Allow, Challenge, Block
5-level challenge escalation

→ Cookie → JS PoW → CAPTCHA → Tarpit → Block

DLP Scanning — 29–71µs

22+ sensitive data patterns
4KB: 29µs / 8KB: 71µs (capped)
Content-type short circuit for binaries

→ ~100µs hidden by running during backend I/O

5-LEVEL CHALLENGE ESCALATION

L1	L2	L3	L4	L5
Cookie	JS PoW	CAPTCHA	Tarpit	Block

LATENCY COMPARISON

SYNAPSE (EDGE)

~450µs

End-to-end proxy with WAF + DLP enabled. All detection runs in-process on the same binary. No network round-trips.

CLOUD WAF

1–2s

Network round-trip to decision backend. DNS resolution, TLS handshake, request forwarding, response inspection — all over the wire.

2,200–4,400× slower

PERFORMANCE BENCHMARKS

SCENARIO	LATENCY	TARGET	STATUS
Fast path (GET, no body)	~300µs	<500µs	✓ Met
Standard (4–8KB body + DLP)	~450µs	<500µs	✓ Met
p95 under load (100 VUs)	<1ms	<1ms	✓ Met

No network hops. No backend calls. All intelligence runs locally at the edge.
Complete request lifecycle in under 500 microseconds.